Where Did You Get Sick?

Place, mobility, time, and uncertainty in public health

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Your homework

How many places did you list? *1–2 · 3–4 · 5 or more*

If a public-health researcher could put you at **one place** only, which did you choose?

- **A** My home
- **B** Where I spent the most time
- **C** Where exposure was most likely
- **D** Another place
- **E** I don't have enough information

0:00–0:02. Hands for the number of places, then for A–E. Say the spread out loud. Listen for two answers that are not on the list. "All of these places matter" is activity space; it comes back with Maya. "It depends on the disease" is right; it comes back with Sam.

What your answer assumes

Your answer What it assumes The follow-up

My home Where you live stands in for what you were exposed to. Convenient, but is it right?

Most time Duration is what matters. Is time alone enough?

Most likely exposure Risk differs from place to place. How would you know?

Not enough information The uncertainty is real. Can a health department wait for it to go away?

0:02–0:04. One row per answer; ask the follow-up before revealing the next row. Home: many health datasets do exactly this. Convenient, and sometimes misleading. Most time: exposure also depends on intensity, on how it happens, and on what actually took place there. Most likely exposure: opens the problem of missing exposure information. Not enough information: this is the wicked problem. Public-health decisions often cannot wait for the uncertainty to disappear.

Same illness, five years later

Who **changed** their answer when the illness appeared five years later?

How confident are you that you found the place of **exposure**?

Very · Somewhat · Not very · Completely uncertain

0:04–0:06. Hands for "changed", then for confidence. Listen for: moving, a new job or school, travel, previous homes, effects that show up later. Then say: "A few minutes ago, one point on a map seemed reasonable. I changed only one thing — the time between exposure and diagnosis — and suddenly that point became much less meaningful." This sets up latency and residential history; Frank brings it back.

What else would you need to know?

PLACE Where else were you?

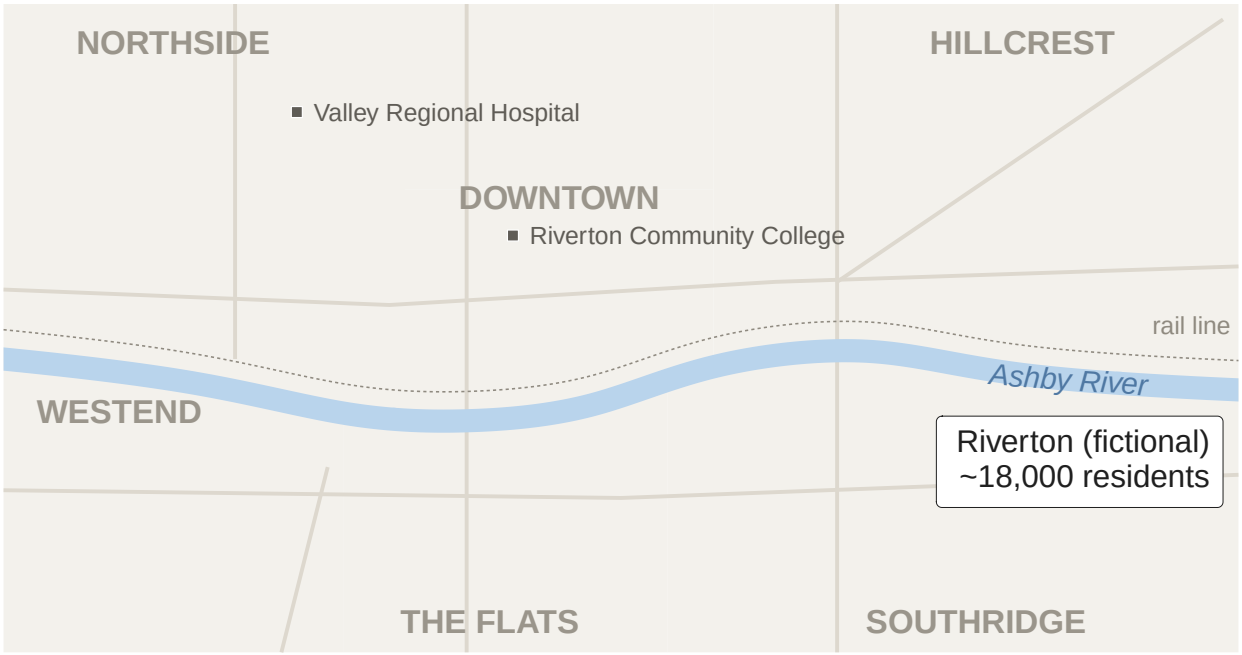
TIME When were you there?

EXPOSURE What happened there?

These are exactly the dimensions that **disappear** when a person is represented only by a home address.

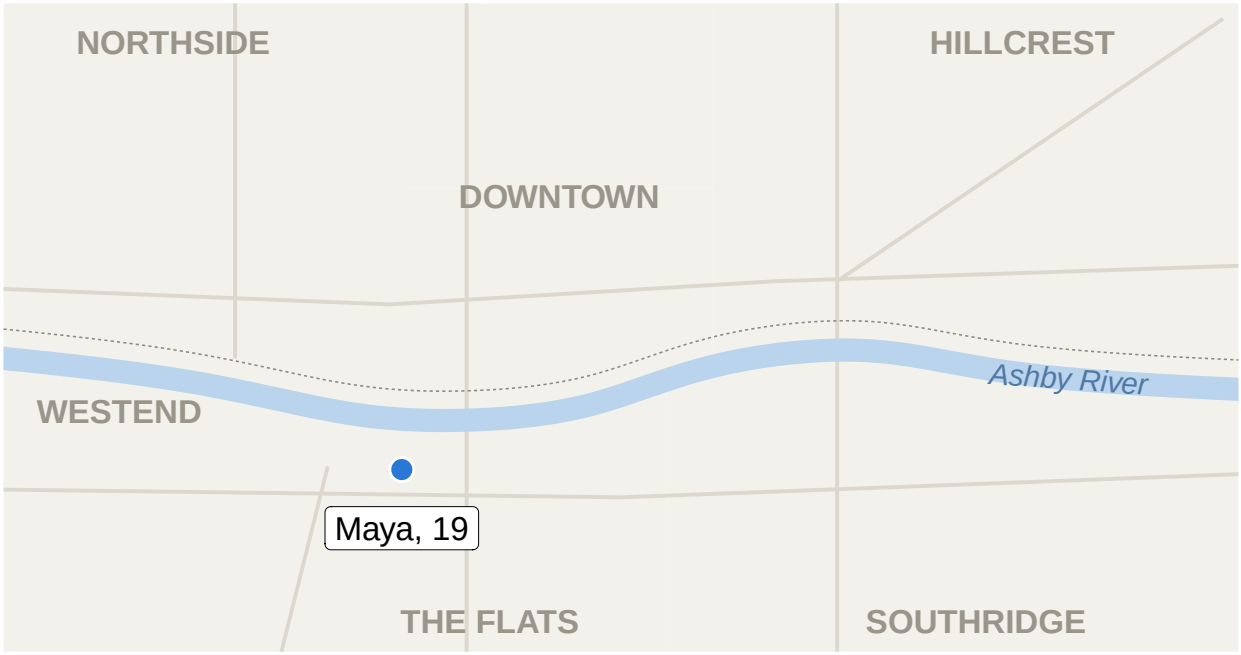
0:06–0:09. Take their requests and sort them into the three boxes out loud, or on the board. Likely: *Place:* work or school, previous homes, travel. *Time:* how long at each place, when symptoms began. *Exposure:* pollution or hazards, food and water, whether others became ill. Disease type and personal risk factors do not fit a box. Park them for Sam and for the smoking question in Riverton. Reveal the key point last.

Meet Riverton



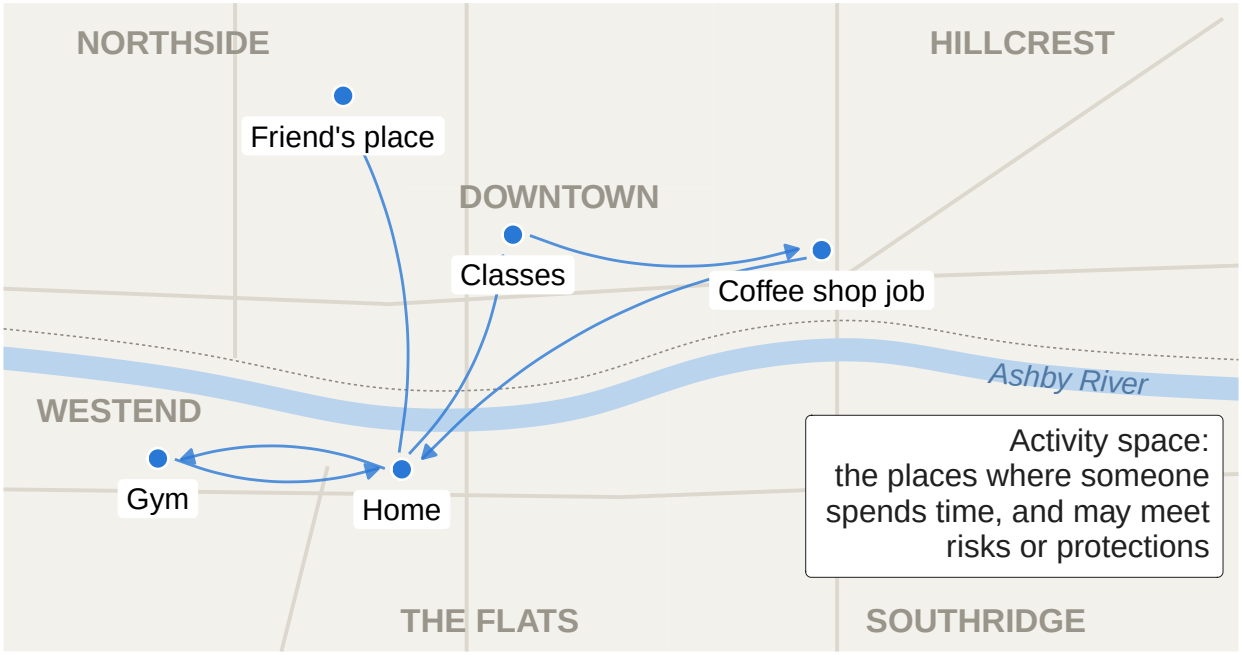
0:09–0:10. Riverton is invented, so nobody's real neighborhood is on trial today. River town, industrial past, about 18,000 people. Point out the Flats (south bank, older rowhouses), Downtown across the bridge, Hillcrest up the hill, and the hospital in Northside.

How a health dataset sees Maya



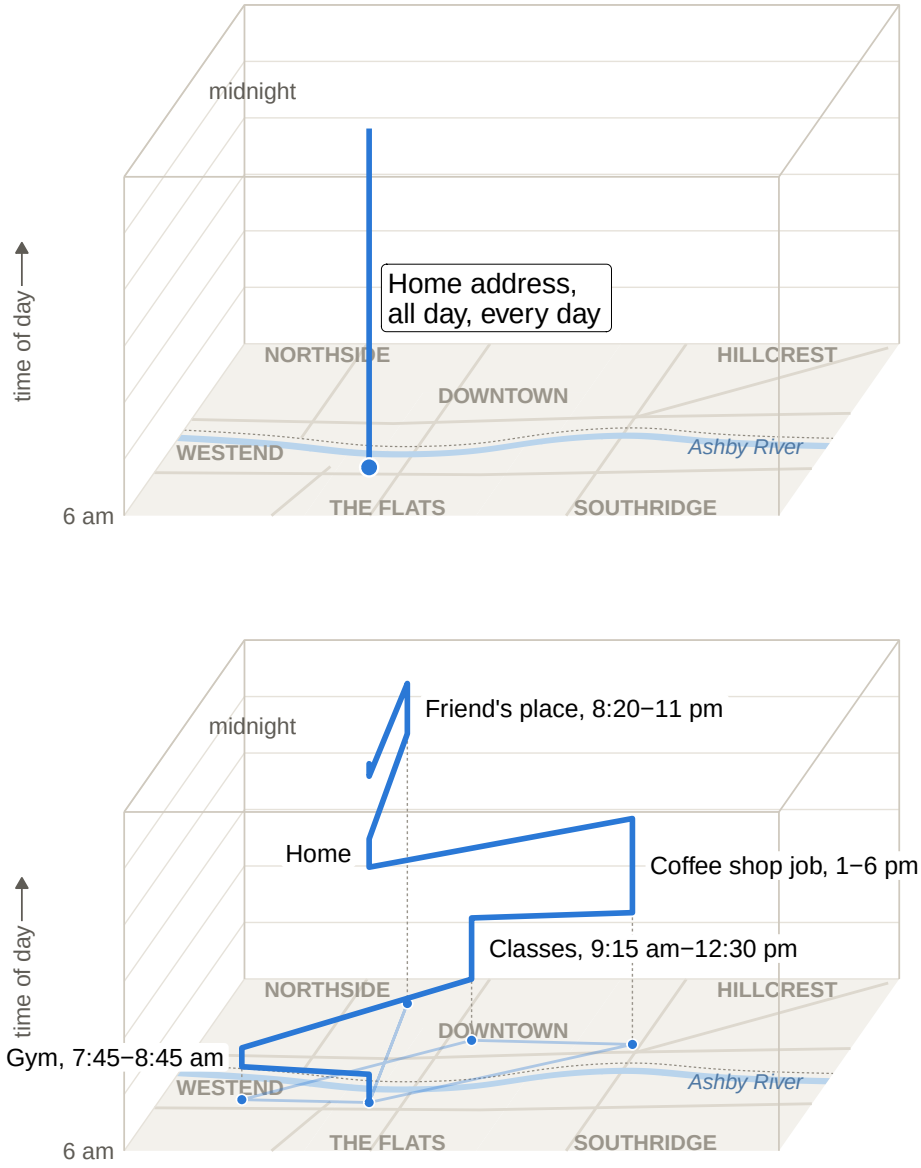
0:10–0:11. "This is how many health datasets see you." Maya, 19, is one geocoded home address in the Flats.

How Maya actually lives



0:11–0:12. "This is closer to how you actually live." Classes downtown, a coffee-shop job in Hillcrest, the gym in Westend, a friend in Northside. Activity space: the places where someone spends time and may meet risks (air, noise, food, other people) or protections (parks, clinics). Ask: **which dot should represent Maya?** Students who wrote "all of these places matter" already had this idea; credit them.

Maya's Thursday as a space-time path



The floor is the map of Riverton; height is the time of day. **Vertical line:** staying in one place. **Sloped line:** moving.

0:12–0:14. First the health record's version: one vertical line at her home address, from 6 am to midnight. Click for her actual Thursday. Read it from the floor up: gym before class, classes downtown, five hours at the coffee shop, home for dinner, a friend's place until 11 pm. Dotted lines drop each stop onto the map. This is a space-time path (Hägerstrand's time geography). The week map showed *where*; this adds *when*. Ask: where and when could Maya have been exposed to something, or exposed someone else? Exposure needs the same place **and** the same time. Sam will need exactly this.

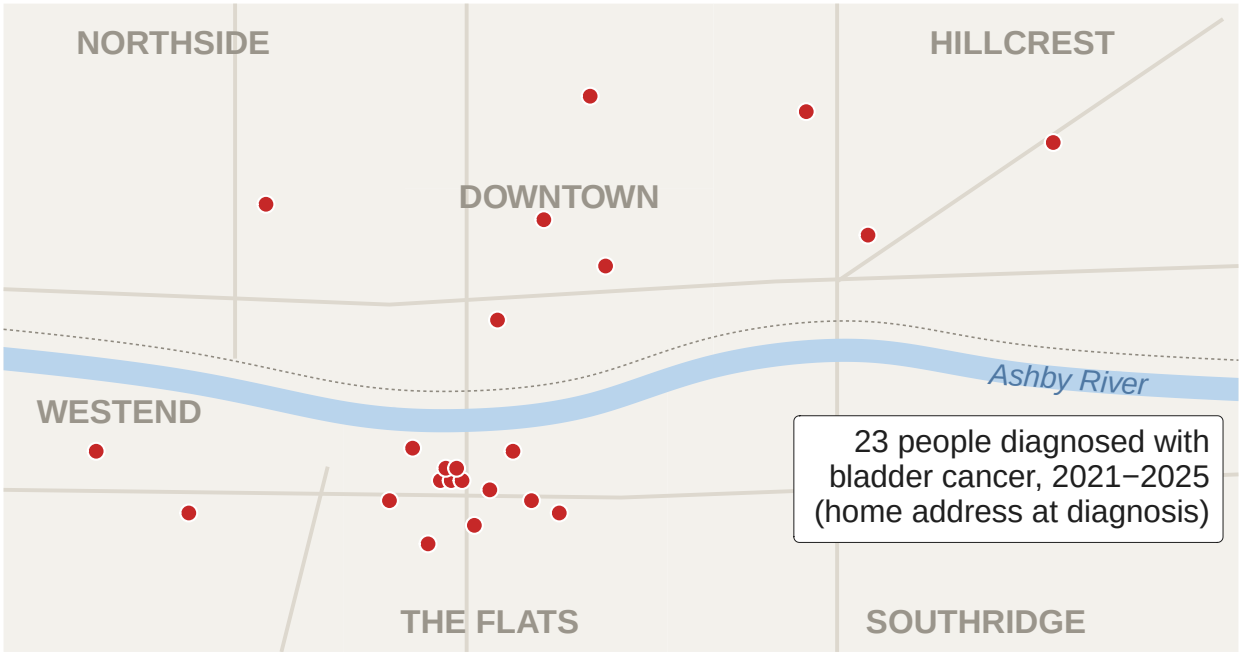
Message 1 of 3

A person is **not a point**.

Health data usually need one location. Your life doesn't have one.

0:14. First of the three messages. Read it and move to the cluster.

What do you think is happening here?



0:14–0:16. Let students call out explanations; don't evaluate yet. Expect: something in the water or soil, a factory, chance, older residents, more testing. Background if asked: smoking causes about half of bladder cancers; certain workplace chemicals (dye, rubber, leather, paint industries) raise risk; about 9 in 10 people diagnosed are over 55.

Before blaming the Flats, what would you want to know?

PLACE Where else did these 23 people live and work?

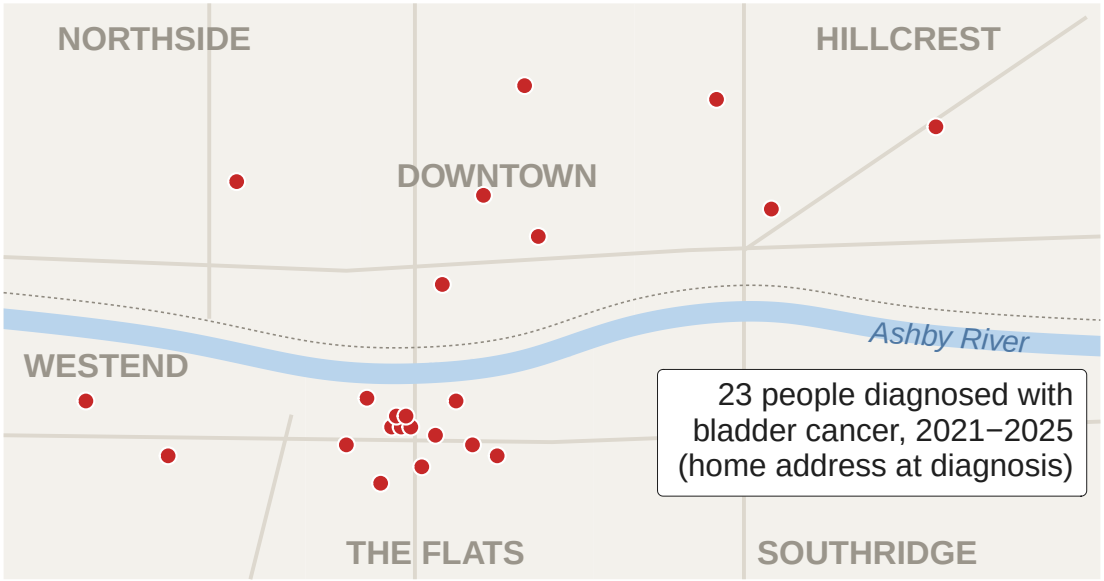
TIME How long have they lived in the Flats? When could exposure have happened?

EXPOSURE Is there a possible source nearby? Who lives in the Flats: age, smoking?

Expected for a town of 18,000 over five years: **about 22**. Riverton's total is ordinary. The **concentration** is not: the Flats holds 16% of residents but 13 of 23 cases.

0:16–0:19. Same three boxes as the homework, now for 23 people. Ask before showing. The expected count uses the national rate (~84,500 new US cases a year, roughly 25 per 100,000) × 18,000 people × 5 years. Point: a cluster can be about *where* rather than *how many*. The next slides answer the boxes one at a time: exposure (the plant), place (jobs), time (who arrived when, and 1985).

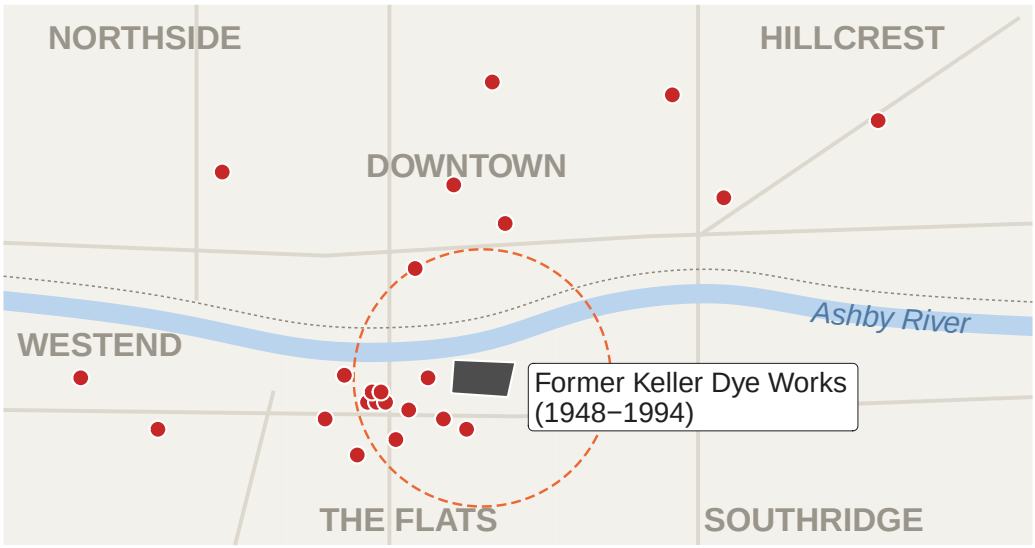
Your explanation, before more evidence?



- **A** Something in the Flats
- **B** Chance
- **C** Who lives there
- **D** Can't tell yet

0:19–0:20. Take hands and remember the split. From here on, the same question after every slide: **has your explanation changed?**

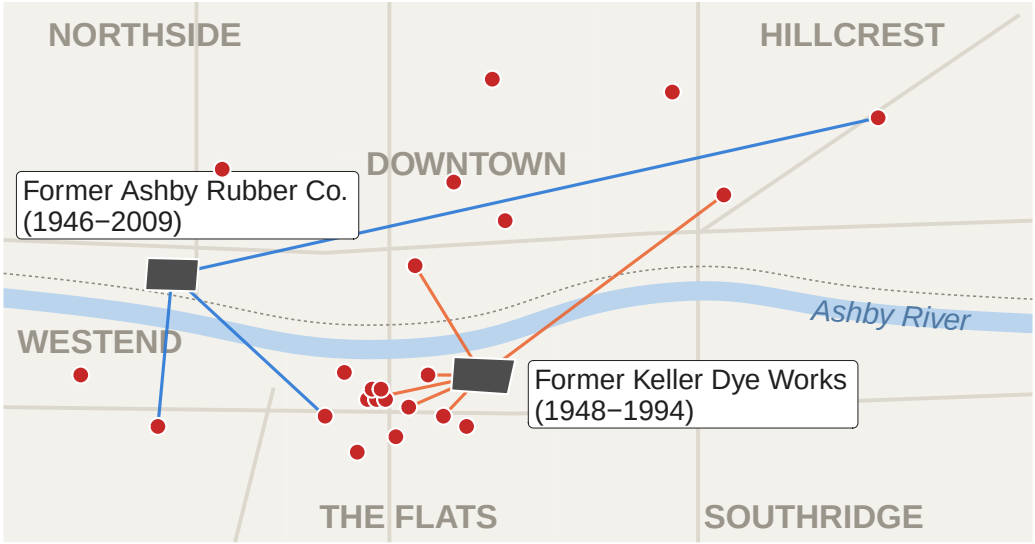
Exposure: a former dye plant next door



11 of 23 cases live within 1 km (dashed circle). Some chemicals used to make dyes cause bladder cancer in exposed workers. **Has your explanation changed?**

0:20–0:22. Keller Dye Works, 1948–1994. Many will now say "it's the plant". Ask how residents (not workers) would have been exposed: air, soil, the river, old wells. "How would you know?" from the homework applies here. Ask: act now, or more evidence first?

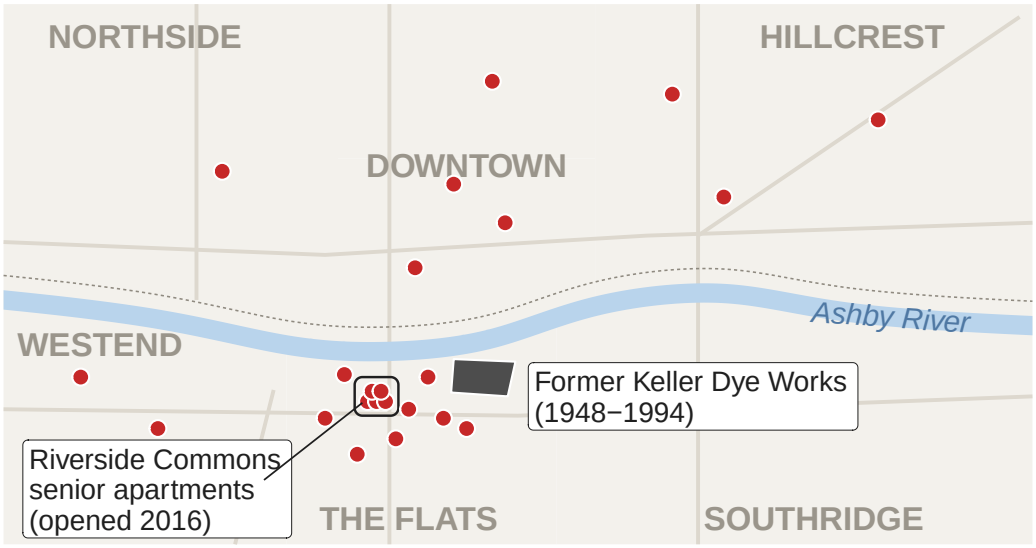
Place: where they worked



6 cases worked at Keller; 3 at Ashby Rubber, which used related chemicals. 4 of these 9 workers live outside the Flats. **Has your explanation changed?**

0:22–0:25. The link may run through the job, not the neighborhood. Some Flats cases are there partly because Keller workers lived near Keller. A second plant across the river now matters too. Which places matter most now?

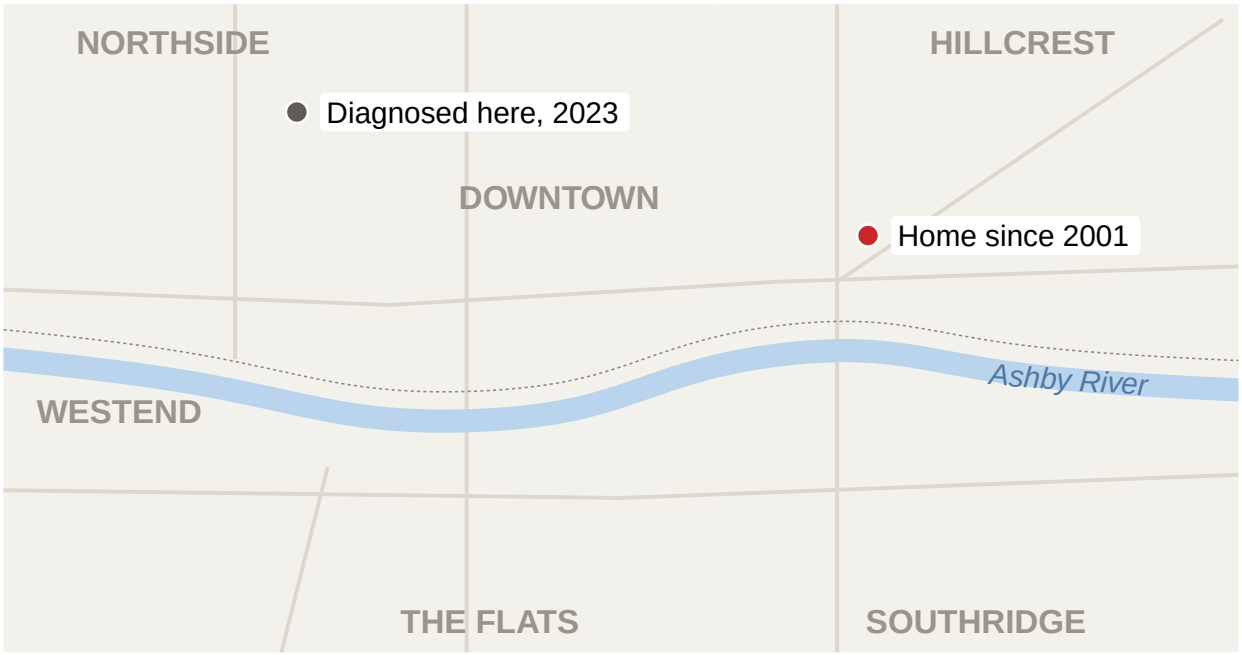
Time: who lives in the Flats now?



5 cases live in Riverside Commons, a senior building opened in 2016; all moved in after age 65. About 9 in 10 people diagnosed with bladder cancer are over 55. **Has your explanation changed?**

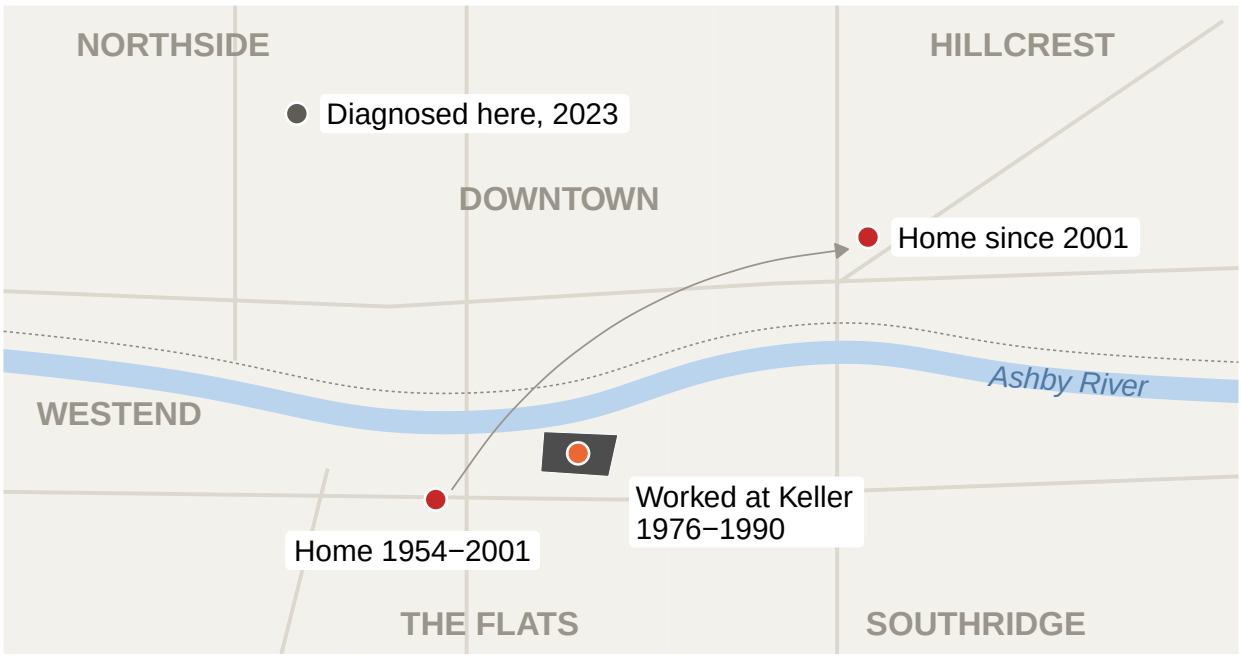
0:25–0:27. A senior apartment building concentrates older people, and older people are more likely to be diagnosed. These five moved in after the plant closed. Part of today's "cluster" is about who lives there, not what happened there.

One of the 23: Frank, 69



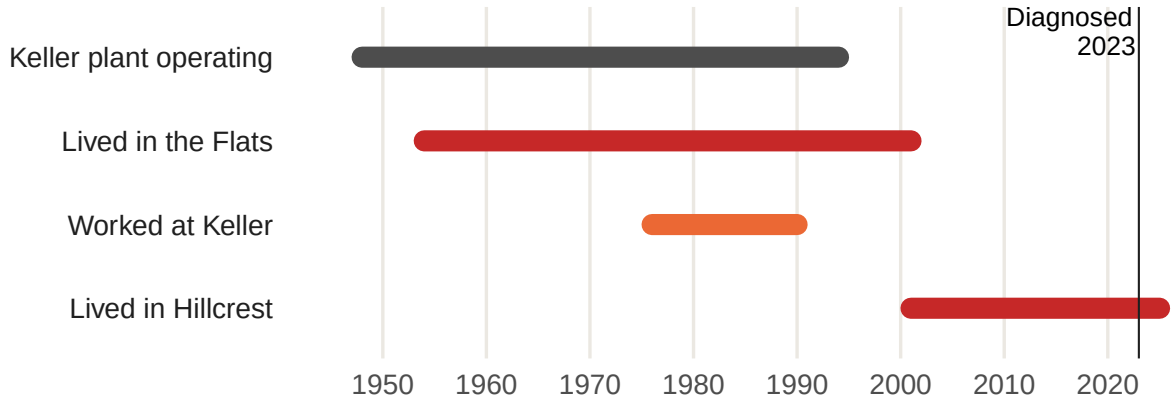
0:27–0:28. On the cluster map Frank is a dot in Hillcrest, far from the Flats. His record also says where he was diagnosed: Valley Regional Hospital. Already two places.

Frank's other places



0:28–0:29. He grew up in the Flats, worked at Keller Dye Works 1976–1990, and moved to Hillcrest in 2001. Nothing on his health record mentions any of this.

Frank's seventy years



Diagnosed in 2023, about thirty years after he last worked at the plant.

0:29–0:30. Call back the homework: "You changed your answers when I added five years. For Frank it is more than thirty." The exposure window, if there was one, closed in 1990; the diagnosis came 33 years later, at a different address.

Where should Frank appear on the map?

A · Where he lives Hillcrest, since 2001

B · Where he was diagnosed Valley Regional Hospital, 2023

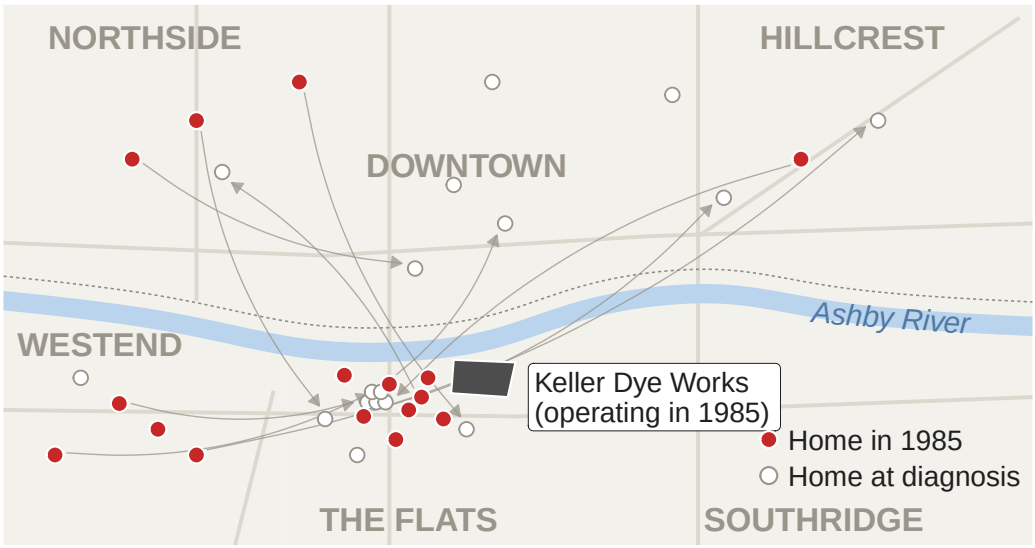
C · Where he may have been exposed The Flats and Keller, 1950s–1990s

- **D** More than one of these
- **E** It depends on the question

For many chronic diseases, exposure and diagnosis are separated by **years or decades**. Some workplace chemicals linked to bladder cancer can take 20 years or more to show an effect.

0:30–0:33. Hands for A–E. Push on E: a map for planning cancer care (where patients live now) differs from a map for finding causes (where they were exposed). Distinguish current residence, place of diagnosis, and place of exposure.

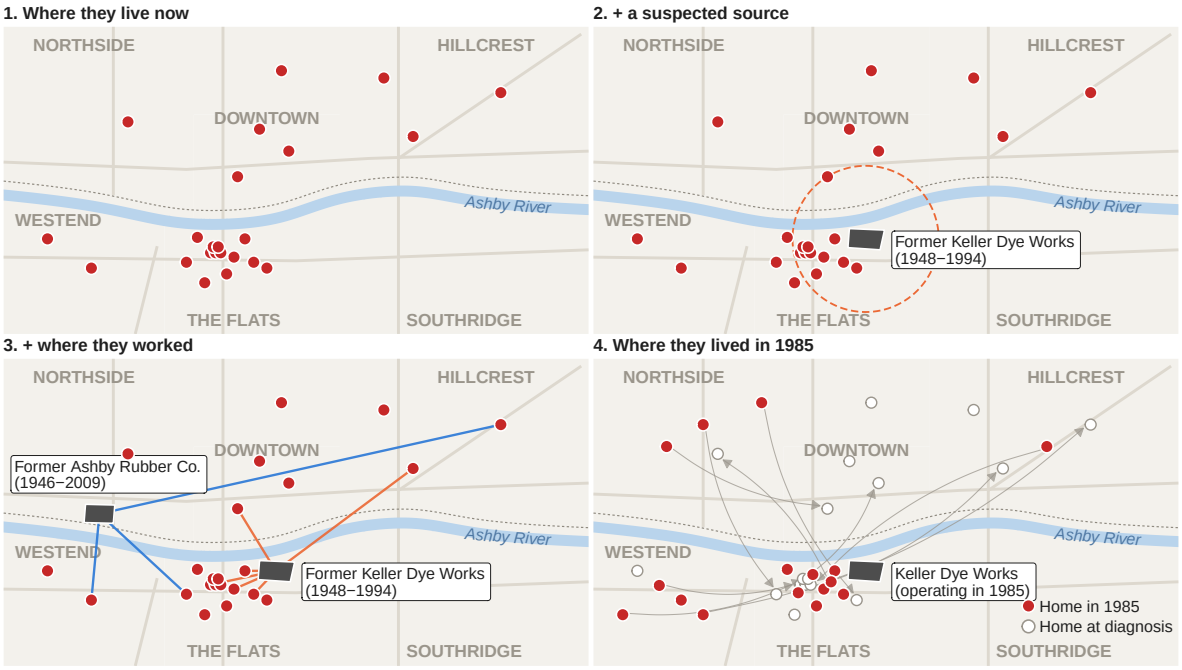
Time: the same 23 people, in 1985



8 lived in the Flats while the plant was running; 3 of them have since moved away. 7 did not live in Riverton at all. **Has your explanation changed?**

0:33–0:36. "Now do for all 23 what we just did for Frank." The cluster is still there in 1985, but it is made of different people. Some of the most plausibly exposed now live elsewhere; some of today's Flats cases arrived recently.

Same cases, four stories



0:36–0:37. Walk the four panels: the Flats → the plant → the jobs → the timing. More information did not make the story simpler; it changed it. About 10 cases still have no identifiable link (smoking, age, chance). There is no single hidden answer.

Change the disease: an infection

Sam, Lehigh sophomore, tests positive on Monday for a contagious respiratory infection (think flu or measles).



Where did Sam get infected? Where might it spread next?

0:37–0:41. For the students who wrote "it depends on the disease": they were right. Same mobility, very different question. For an infection we look back days, not decades, and we also look forward: Sam's lecture halls, coworkers in Allentown, the gym in Easton, the bus to Philadelphia. Health departments do this in practice: for measles they publish the places and times where people may have been exposed. Here mobility is both a clue and the pathway.

Same mobility, different questions

	Chronic disease (Frank)	Infection (Sam)
Time window	Years to decades	Days to weeks
Direction	Look backward	Look backward **and** forward
Mobility tells us	Where exposure *might* have happened	Where it happened, and where it may go

0:41–0:42. Quick slide. Skip it if time is short.

Message 2 of 3

Where disease is **diagnosed** is not necessarily where it was **acquired**.

For Frank: decades earlier, across town. For Sam: days earlier, in another city.

0:42–0:43. Second message. Then back to Riverton for the decision.

You are the Riverton Health Department

Possible actions

- Test soil, air, and water at the former plant
- Contact former Keller and Ashby workers
- Offer screening
- Collect residential and work histories
- Issue a public warning
- Restrict access to the site
- Keep monitoring

Your call?

- **A** Act now
- **B** Targeted investigation
- **C** Collect more data first
- **D** Routine surveillance

0:43–0:48. Hands for A–D, then ask one student from each camp to defend it. Useful complication: screening people without symptoms for bladder cancer is not routinely recommended; it can produce false alarms and unnecessary procedures. Who should decide: scientists, health officials, elected leaders, residents?

Acting too early, acting too late

Too early

- Fear and stigma for the Flats
- Falling home values; worried landlords and seniors
- Money spent on the wrong problem
- Trust lost if the threat wasn't real

Too late

- Preventable illness continues
- Former workers and families never warned
- Trust lost if officials knew and waited

How certain should we be before taking action? And how much about people's movements should a health department be allowed to collect?

0:48–0:51. Is waiting for better evidence always the most scientific choice? Privacy: residential and work histories, maybe phone location data. Useful for public health; intrusive for individuals.

Better questions

Where did you get sick?

- Where **could** exposure have occurred?
- **When?**
- **Who else** might have been exposed?
- What evidence would **change** our interpretation?
- When do we know **enough to act**?

0:51–0:53. Replace the opening question with these.

Three messages

1. A person is **not a point**.
2. Where disease is **diagnosed** is not necessarily where it was **acquired**.
3. Decisions still have to be made **under uncertainty**.

In recitation, you are the Riverton Health Department.

0:53–0:55. Return to the homework: your answers disagreed, and several of them were reasonable. That disagreement is why locating exposure is a real public-health problem, not a failure to find the right answer. Remaining time: questions.